

# Introducción a la Inteligencia Artificial

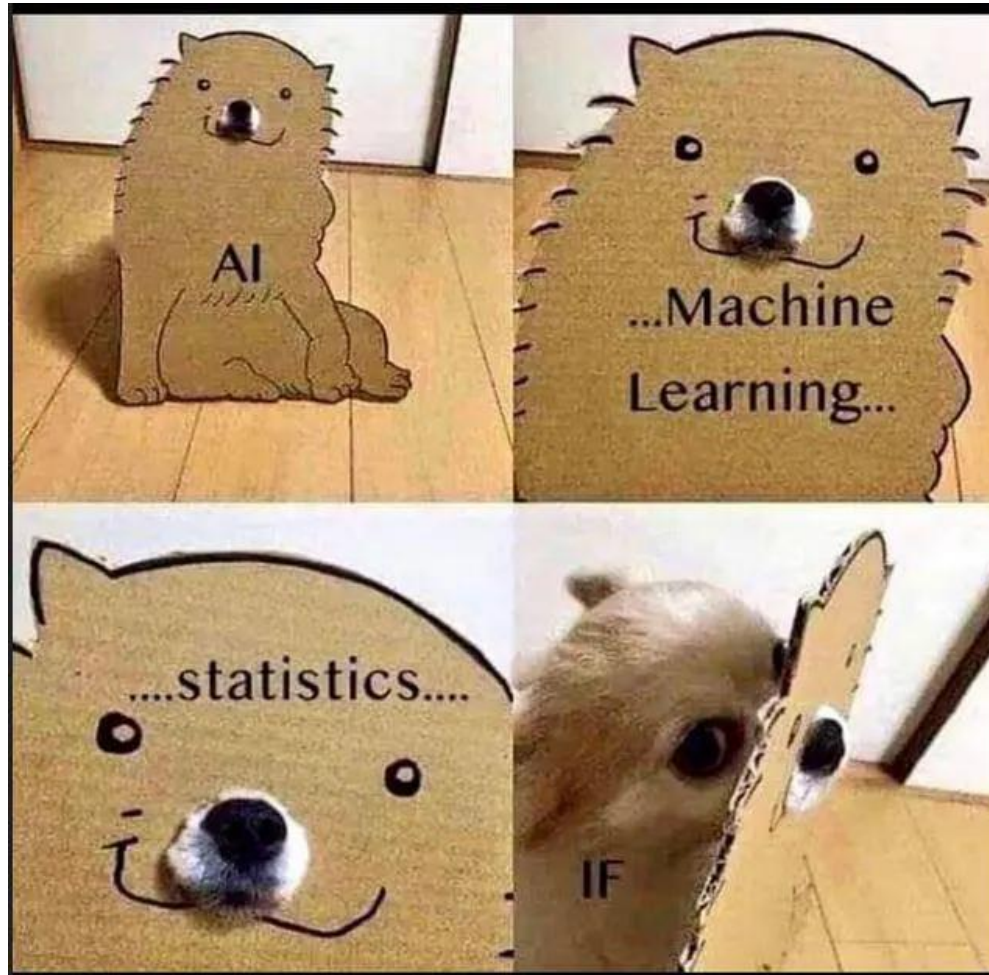
Daniel Barajas Higuera

MLEARN



# Machine Learning

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# ¿Las maquinas pueden pensar?

VOL. LIX. No. 236.]

[October, 1950

## MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

### I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. TURING

#### 1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning



“  
The original  
question, 'Can  
machines  
think?' I believe  
to be too  
meaningless to  
deserve  
discussion

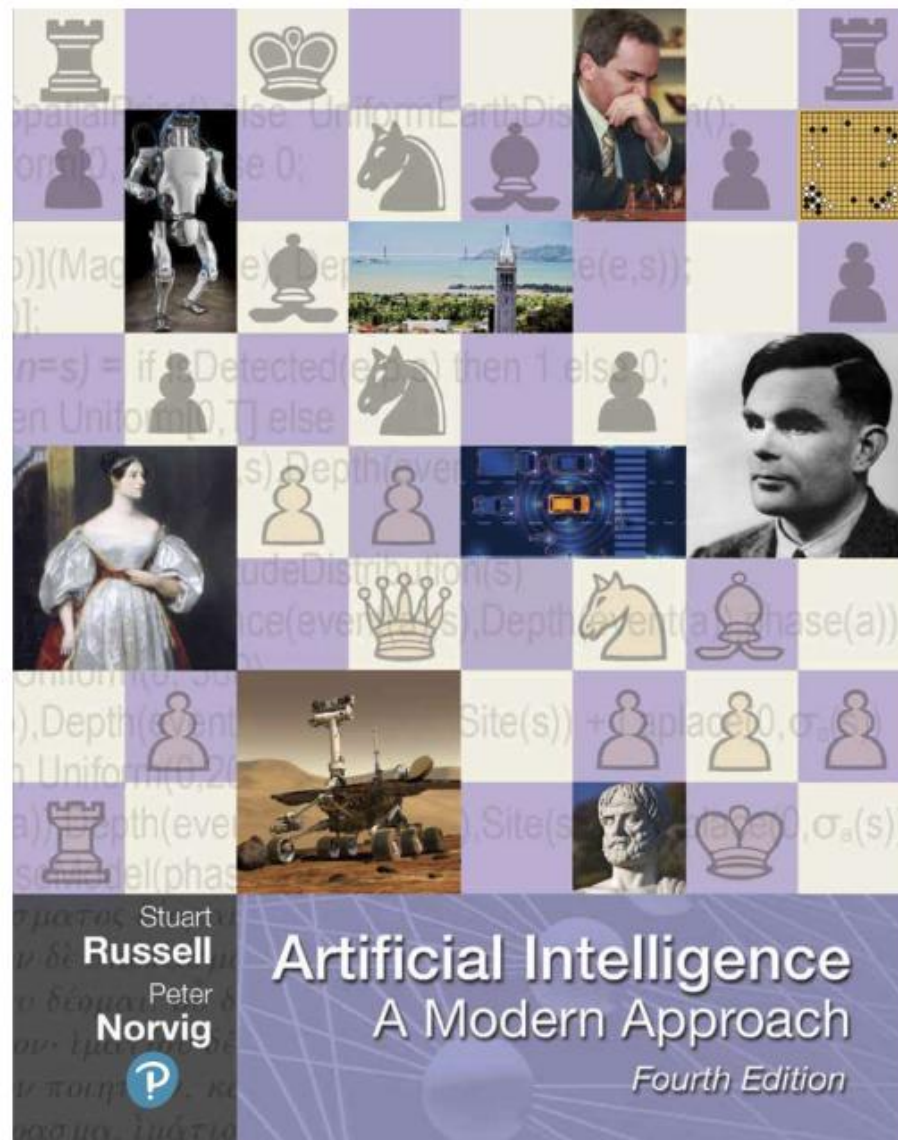
”

- Alan Turing

# Inteligencia Artificial



# Inteligencia Artificial



# Inteligencia Artificial

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# Conocimiento incierto y razonamiento

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- 24.6 State of the art . . . . .
- Summary . . . . .
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### 25 Computer Vision

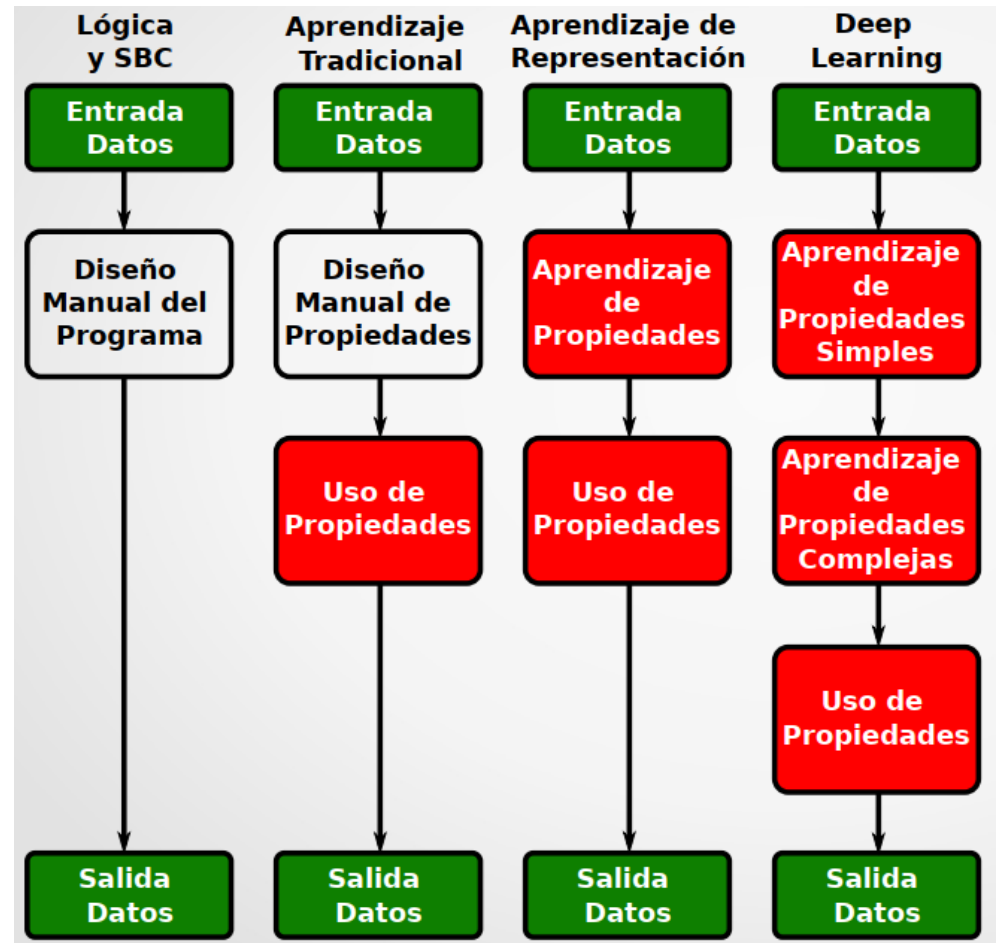
- 25.1 Introduction . . . . .
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- Summary . . . . .
- Bibliographical and Historical Notes . . . . .

# Enfoques de Aprendizaje en Inteligencia Artificial

- 80's: Lógica y Sistemas Basados en Conocimiento.
- 90's: Probabilidad y Refuerzo.
- 00's: Geométrico.
- 10's: Redes Neuronales.



# Enfoques de Aprendizaje en Inteligencia Artificial

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## Enfoques de Aprendizaje

- **Enfoque Búsqueda en Grafos:**  
Aprendizaje por Refuerzo (MDP).
- **Enfoque Probabilístico:**  
Árboles de Decisión.  
Redes Bayesianas.
- **Enfoque Lógico:**  
EV, Aq, FOIL.
- **Enfoque Geométrico:**  
k-NN, k-Means, SVM (distancias).
- **Enfoque Conexionista:**  
redes neuronales.

# Definiciones

Machine learning  $\subseteq$  artificial intelligence

## ARTIFICIAL INTELLIGENCE

Design an intelligent agent that perceives its environment and makes decisions to maximize chances of achieving its goal.  
Subfields: vision, robotics, machine learning, natural language processing, planning, ...

## MACHINE LEARNING

Gives "computers the ability to learn without being explicitly programmed" (Arthur Samuel, 1959)

### SUPERVISED LEARNING

Classification, regression

### UNSUPERVISED LEARNING

Clustering, dimensionality  
reduction, recommendation

### REINFORCEMENT LEARNING

Reward maximization

*Machine Learning for Humans* 🤖🧠

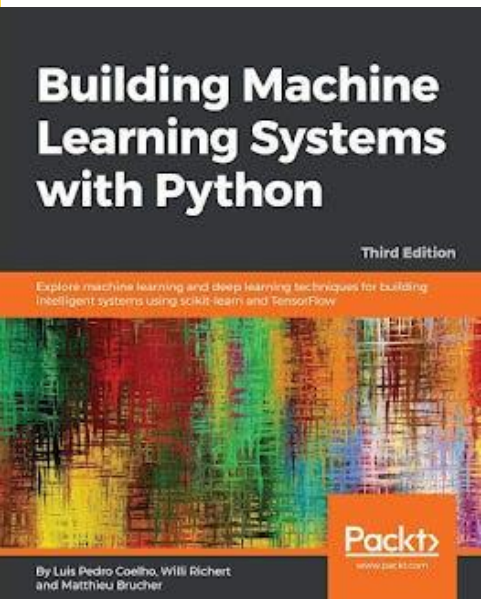
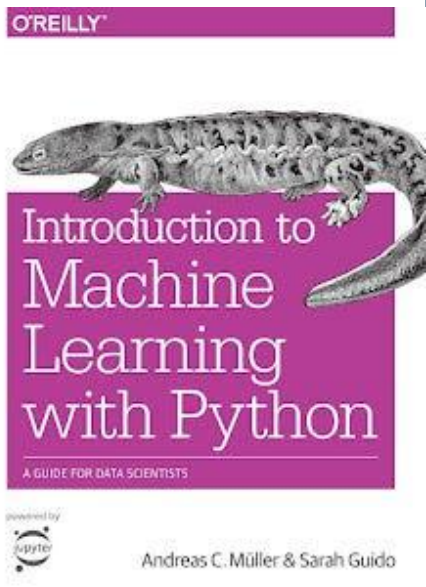
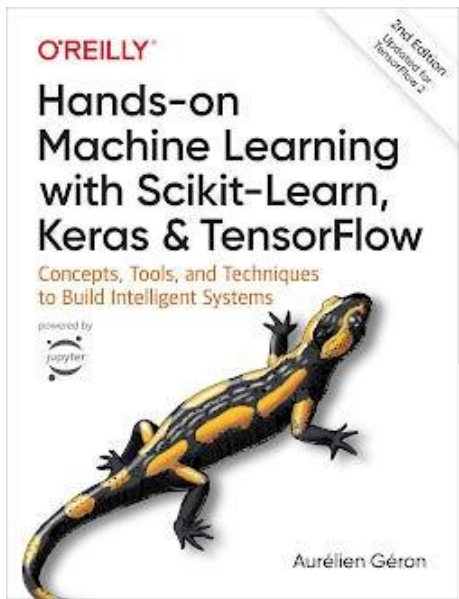
# Machine Learning y Deep Learning

|   | Tema                                                                                                    | Ejemplo de Aplicación                                                   |
|---|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | Introducción a Machine Learning y conceptos básicos.                                                    | Clasificación de correos electrónicos como spam o no spam.              |
| 2 | Matemáticas Fundamentales para Machine Learning: Álgebra lineal y Cálculo.                              | Predicción de precios de casas utilizando regresión lineal.             |
| 3 | Preprocesamiento de Datos para Machine Learning: Limpieza, manipulación y selección de características. | Análisis de sentimiento en redes sociales.                              |
| 4 | Aprendizaje Supervisado: Regresión lineal y no lineal.                                                  | Predicción de ventas de productos basada en características históricas. |
| 5 | Aprendizaje Supervisado: Clasificación binaria y multiclase.                                            | Clasificación de enfermedades a partir de datos médicos.                |
| 6 | Aprendizaje No Supervisado: Agrupamiento (clustering) y Detección de anomalías.                         | Segmentación de clientes para campañas de marketing personalizadas.     |
| 7 | Aprendizaje No Supervisado: Reducción de dimensionalidad.                                               | Visualización de datos en alta dimensión para análisis exploratorio.    |

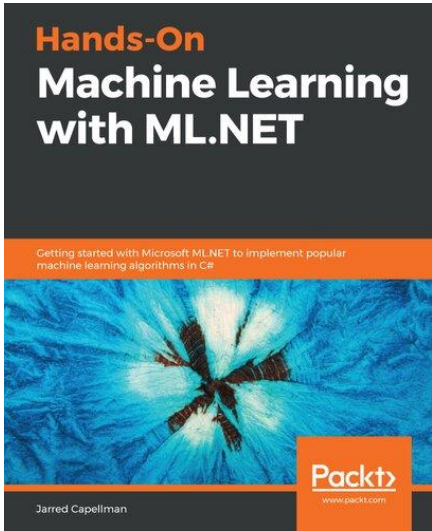
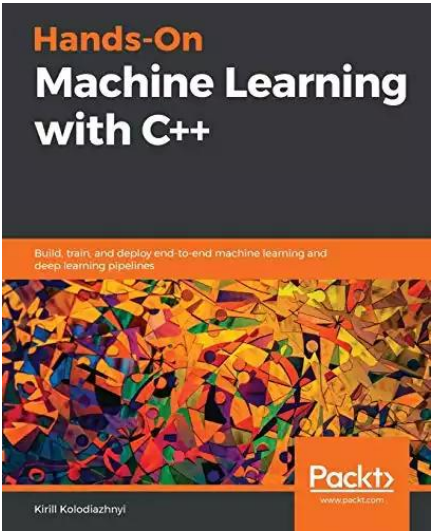
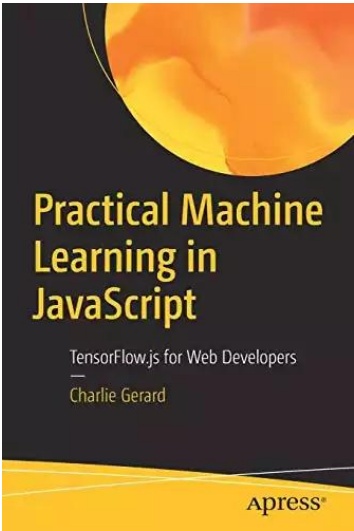
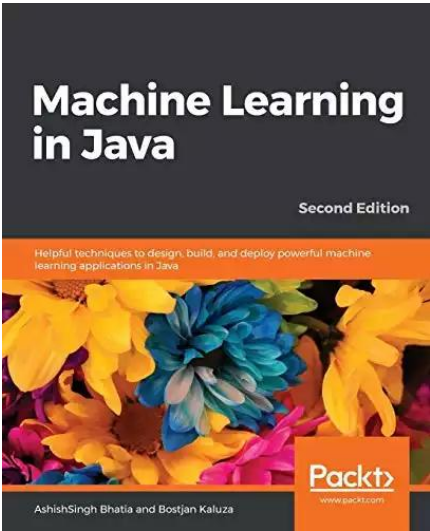
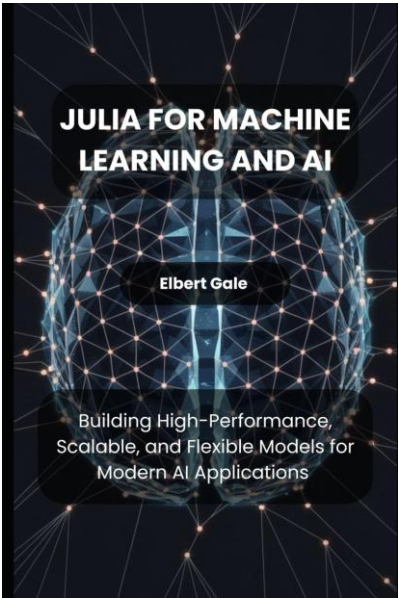
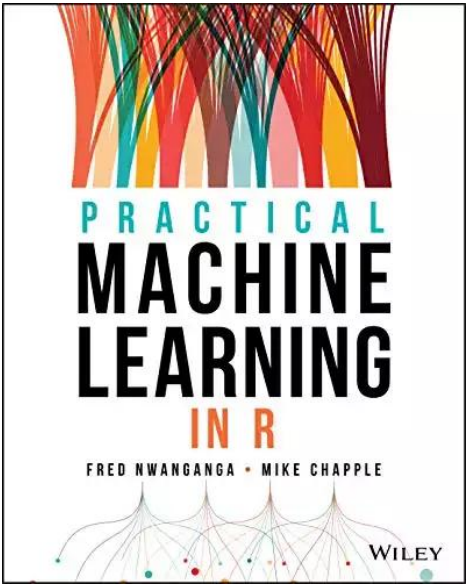
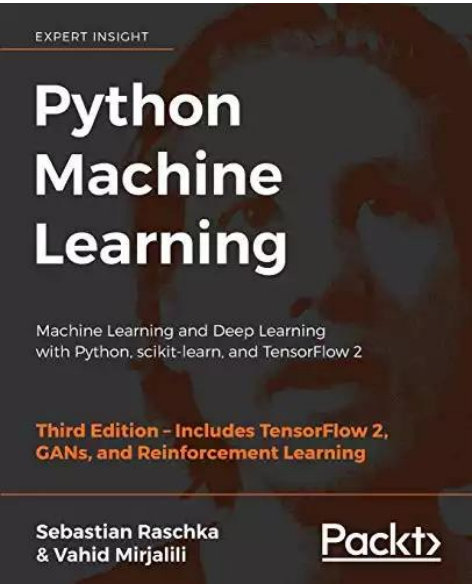
# Machine Learning y Deep Learning

|    | Tema                                                                                                                          | Ejemplo de Aplicación                                                      |
|----|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 8  | Aprendizaje Profundo (Deep Learning): Redes neuronales artificiales.                                                          | Reconocimiento de dígitos escritos a mano (MNIST dataset).                 |
| 9  | Aprendizaje Profundo (Deep Learning): Arquitecturas populares (por ejemplo, CNN y RNN).                                       | Clasificación de imágenes médicas para detección de enfermedades.          |
| 10 | Técnicas Avanzadas en Machine Learning: Aprendizaje semi-supervisado y Aprendizaje por refuerzo.                              | Control de robots en entornos complejos mediante aprendizaje por refuerzo. |
| 11 | Técnicas Avanzadas en Machine Learning: Métodos ensemble (por ejemplo, Random Forest y Gradient Boosting).                    | Predicción de churn en una empresa de telecomunicaciones.                  |
| 12 | Consideraciones Éticas y de Interpretabilidad en Machine Learning.                                                            | Identificación y mitigación de sesgos en algoritmos de contratación.       |
|    | Proyecto Final: Desarrollo de un proyecto aplicando técnicas de Machine Learning a datos reales y presentación de resultados. |                                                                            |

# Libros



# Libros



# Librerías

| Tema                                                                                                           | Python                                    | Julia                                     | R                                        | C++                    | C#                         | Java                              | JavaScript             |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|------------------------------------------|------------------------|----------------------------|-----------------------------------|------------------------|
| <b>Introducción a Machine Learning y conceptos básicos.</b>                                                    |                                           |                                           |                                          |                        |                            |                                   |                        |
| <b>Matemáticas Fundamentales para Machine Learning: Álgebra lineal y Cálculo.</b>                              | NumPy<br>SciPy<br>SymPy                   | LinearAlgebra<br>Calculus<br>SymPy        | Matrix<br>pracma<br>calculus<br>numDeriv | Eigen                  |                            | Apache<br>Commons<br>Math         | mathjs                 |
| <b>Preprocesamiento de Datos para Machine Learning: Limpieza, manipulación y selección de características.</b> | Pandas<br>NumPy<br>scikit-learn           | DataFrames<br>CSV<br>Query                | tidyverse<br>dplyr<br>datatable          | Boost C++<br>Libraries | MLNET<br>AccordNET         | Weka<br>Apache<br>Commons<br>Math | DataForge<br>Pandas-js |
| <b>Aprendizaje Supervisado: Regresión lineal y no lineal.</b>                                                  | scikit-learn<br>statsmodels               | GLM<br>DecisionTree<br>RandomForest       | stats<br>glmnet<br>randomForest          | MLPack                 | MLNET<br>AccordNET         | Weka<br>Apache<br>Commons<br>Math | ml-regression          |
| <b>Aprendizaje Supervisado: Clasificación binaria y multiclase.</b>                                            | scikit-learn<br>XGBoost<br>LightGBM       | MLJ<br>XGBoost<br>DecisionTree            | caret<br>e1071<br>randomForest           | MLPack<br>Dlib         | MLNET<br>AccordNET         | Weka<br>Apache<br>Commons<br>Math | ml-classify            |
| <b>Aprendizaje No Supervisado: Agrupamiento (clustering) y Detección de anomalías.</b>                         | scikit-learn<br>KMeans<br>IsolationForest | Clustering<br>DBSCAN<br>OutliersDetection | cluster<br>kmeans<br>dbscan              | MLPack<br>Dlib         | AccordNET<br>ClusteringNET | Weka<br>Apache<br>Commons<br>Math | ml-kmeans              |

# Librerías

| Tema                                                                                                       | Python                                   | Julia                                                | R                               | C++                   | C#                                              | Java                               | JavaScript       |
|------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------|---------------------------------|-----------------------|-------------------------------------------------|------------------------------------|------------------|
| Aprendizaje No Supervisado: Reducción de dimensionalidad.                                                  | scikit-learn<br>PCA<br>UMAP              | MultivariateStats<br>PCA<br>Dimensionality Reduction | prcomp<br>factoextra<br>dplyr   | MLPack                | AccordNET<br>Principal Component Analysis (PCA) | Weka<br>Apache Commons Math        | ml-pca           |
| Aprendizaje Profundo (Deep Learning): Redes neuronales artificiales.                                       | TensorFlow<br>Keras<br>PyTorch           | Flux<br>Knet<br>TensorFlow                           | keras<br>tensorflow             | TensorFlow C++<br>API | MLNET<br>AccordNET                              | DL4J<br>(DeepLearning4j)<br>Tribuo | TensorFlowjs     |
| Aprendizaje Profundo (Deep Learning): Arquitecturas populares (por ejemplo, CNN y RNN).                    | TensorFlow<br>Keras<br>PyTorch<br>NLTK   | Flux<br>Knet<br>TextAnalysis                         | keras<br>tensorflow<br>text2vec | TensorFlow C++<br>API | MLNET<br>AccordNET                              | DL4J<br>(DeepLearning4j)<br>Tribuo | TensorFlowjs     |
| Técnicas Avanzadas en Machine Learning: Aprendizaje semi-supervisado y Aprendizaje por refuerzo.           | scikit-learn<br>TensorFlow<br>OpenAI Gym | MLJ<br>TensorFlow<br>Reinforcement Learning          | caret<br>h2o<br>rlang           | MLPack                | MLNET<br>AccordNET                              | Weka<br>Apache Commons Math        |                  |
| Técnicas Avanzadas en Machine Learning: Métodos ensemble (por ejemplo, Random Forest y Gradient Boosting). | scikit-learn<br>XGBoost<br>LightGBM      | DecisionTree<br>XGBoost<br>GradientBoosting          | randomForest<br>xgboost<br>gbm  | MLPack<br>Dlib        | MLNET<br>AccordNET                              | Weka<br>Apache Commons Math        | ml-random-forest |
| Consideraciones Éticas y de Interpretabilidad en Machine Learning.                                         | fairlearn<br>shap                        | Fairness<br>SHAP                                     | fairness<br>interpretable       |                       |                                                 |                                    |                  |

# Inteligencia Artificial

